

Math 161

Problem set 4 solutions

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1. Suppose that $\lim_{x \rightarrow a} f(x) = L$, so we want to show that $\lim_{h \rightarrow 0} f(a+h) = L$. Given $\epsilon > 0$, there exists $\delta > 0$ such that, for any $x \in \mathbb{R}$, if $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$. Now for any $h \in \mathbb{R}$ let $x := a + h$. If $0 < |h| < \delta$, or equivalently $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$, i.e. $|f(a+h) - L| < \epsilon$. Thus $\lim_{h \rightarrow 0} f(a+h) = L$ as desired.

Conversely, assume that $\lim_{h \rightarrow 0} f(a+h) = L$. Given $\epsilon > 0$, there exists $\delta > 0$ such that, for any $h \in \mathbb{R}$, if $0 < |h| < \delta$ then $|f(a+h) - L| < \epsilon$. For any $x \in \mathbb{R}$, we put $h := x - a$. If $0 < |x - a| < \delta$, or equivalently $0 < |h| < \delta$, then $|f(a+h) - L| < \epsilon$, i.e. $|f(x) - L| < \epsilon$. This means that $\lim_{x \rightarrow a} f(x) = L$.

2. If $\lim_{x \rightarrow a} f(x) = L$, then there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $|f(x) - L| < 1$ (it is not important here that we take $\epsilon = 1$, only that we make ϵ some constant). Thus $0 < |x - a| < \delta$ implies that

$$|f(x)| - |L| \leq |f(x) - L| < 1$$

and hence $|f(x)| < |L| + 1$. So we have proved the claim with $M = |L| + 1$.

3. We apply the previous problem. The contrapositive of that statement says that if, for any $\delta > 0$ and any $M > 0$, there exists $x \in \mathbb{R}$ such that $0 < |x - a| < \delta$ and $|f(x)| \geq M$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

Given $\delta > 0$ and $M > 0$, we take $x := \min(\frac{\delta}{2}, \frac{1}{M})$. Then $0 < x < \frac{\delta}{2}$, and hence $0 < |x| < \frac{\delta}{2}$ because $x > 0$ implies that $|x| = x$. On the other hand, since $x \leq \frac{1}{M}$, we have $\frac{1}{x} \geq M$ and hence

$$|\frac{1}{x}| = \frac{1}{x} \geq M.$$

Thus $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist by problem 2.

One can also show that $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist by using the definition of the limit directly.

4. (a) This is false. Namely, take $f(x) = \frac{1}{x}$ and $g(x) = -\frac{1}{x}$. We know that $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist by problem 3. We claim that $\lim_{x \rightarrow 0} -\frac{1}{x}$ also does not exist. If it did exist, then by part (ii) of the theorem on limits we would have

$$\lim_{x \rightarrow 0} \frac{1}{x} = \lim_{x \rightarrow 0} (-1) \left(-\frac{1}{x}\right) = \left(\lim_{x \rightarrow 0} -1\right) \left(\lim_{x \rightarrow 0} -\frac{1}{x}\right) = - \left(\lim_{x \rightarrow 0} -\frac{1}{x}\right),$$

which contradicts problem 3.

Despite this, the limit

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} + \left(-\frac{1}{x}\right)\right) = \lim_{x \rightarrow 0} 0 = 0$$

does exist.

- (b) This is true. Namely, if $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} (f(x) + g(x))$ exist, then by part (ii) of the theorem on limits we have

$$\lim_{x \rightarrow a} (-f(x)) = \left(\lim_{x \rightarrow a} -1\right) \left(\lim_{x \rightarrow a} f(x)\right) = - \lim_{x \rightarrow a} f(x),$$

and hence by part (i) we have

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} (f(x) + g(x) - f(x)) = \lim_{x \rightarrow a} (f(x) + g(x)) + \lim_{x \rightarrow a} (-f(x)) = \lim_{x \rightarrow a} (f(x) + g(x)) - \lim_{x \rightarrow a} f(x).$$

(c) This is false. Take $f(x) = x$ and $g(x) = \frac{1}{x}$. Then $\lim_{x \rightarrow 0} x = 0$, and

$$\lim_{x \rightarrow 0} x \cdot \frac{1}{x} = \lim_{x \rightarrow 0} 1 = 1$$

because $\frac{x}{x} = 1$ for all $x \neq 0$. Nonetheless, we know that $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist by problem 3.

5. By hypothesis, we have

$$\lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} h(x)$$

for some $L \in \mathbb{R}$. We would like to show that $\lim_{x \rightarrow a} g(x) = L$, i.e. for any $\epsilon > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $|g(x) - L| < \epsilon$. By hypothesis there exist $\delta_1, \delta_2 > 0$ such that $0 < |x - a| < \delta_1$ implies that $|f(x) - L| < \epsilon$ and $0 < |x - a| < \delta_2$ implies that $|h(x) - L| < \epsilon$. Let $\delta := \min(\delta_1, \delta_2)$, whence $0 < |x - a| < \delta$ implies that $|f(x) - L| < \epsilon$ and $|h(x) - L| < \epsilon$. The former implies that $-\epsilon < f(x) - L$, from which we deduce that

$$-\epsilon < f(x) - L \leq g(x) - L.$$

Similarly, from the inequality $|h(x) - L| < \epsilon$ we conclude that

$$g(x) - L \leq h(x) - L < \epsilon.$$

Combining the above yields $-\epsilon < h(x) - L < \epsilon$, i.e. $|h(x) - L| < \epsilon$.